让蚂蚁变僵尸的真菌

The Fungus That Turns Ants into Zombies



蚂蚁为何会咬住叶片不松口?

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RECALL 回想巩固

英文原文

Deep in a Thai rainforest, a carpenter ant breaks from its colony's column in the lofty canopy. Swaying on unsteady legs, it wanders downward, as if intoxicated, into leafy undergrowth. Around midday it clenches its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

By dusk the insect is lifeless, yet the bite holds rigidly. A fungus has destroyed its jaw muscles, so they cannot loosen. Two or three days later a fungal stalk pushes from its head and releases infectious spores on to the ground below, where foraging ants tread.

The invader, Ophiocordyceps unilateralis, is fussy about location. David Hughes and colleagues found the bites mostly on the north-west side of plants, where the air is humid and the temperature suits the fungus. When researchers moved infected ants higher, the fungus grew abnormally. Placed on the ground, the bodies disappeared.

In 2017 a Penn State team examined infected ants with three-dimensional microscope imaging. Fungal cells had formed an intertwined lattice around muscle fibres in the legs and jaw, a picture that evokes the strings of a puppet. Yet the fungus had not infiltrated the brain.

A Planet Earth episode, with David Attenborough's narration, showed one such ant. It helped inspire The Last of Us, a 2013 video game whose makers envisioned the parasite jumping to humans. The hypothetical scenario gave gamers a fictional nightmare.

The real parasite's roots run far deeper, into the geological record. A 48-million-year-old fossil leaf from Messel, in Germany, bears the same characteristic bite scars.

中文译文

在泰国雨林深处，一只木匠蚁脱离了高处树冠中的蚁群队列。它腿脚不稳，摇摇晃晃地向下游荡，仿佛喝醉一般，钻进枝叶茂密的林下植被。正午前后，它用双颚紧紧咬住一片叶子背面的主叶脉，那里离地面约25厘米。

到黄昏时，蚂蚁已经死去，却仍牢牢咬住叶子。真菌破坏了它的颚部肌肉，使其无法放松。两三天后，一根菌柄从蚂蚁头部伸出，向下方地面散播具有传染性的孢子，那里正有觅食的蚂蚁经过。

这种入侵者叫偏侧蛇虫草菌，对位置很挑剔。戴维·休斯及其同事发现，咬合点大多位于植物的西北侧，那里空气潮湿，温度也适合真菌。研究者将受感染的蚂蚁移到更高处后，真菌生长得不正常；放到地面的蚁尸则不见了。

2017年，宾夕法尼亚州立大学的团队通过三维显微成像检查受感染的蚂蚁。真菌细胞在腿部和颚部的肌肉纤维周围织成相互交错的网络，令人联想到操纵木偶的提线。然而，真菌并未侵入蚂蚁的大脑。

由戴维·阿滕伯勒解说的纪录片《地球脉动》曾在一集中展示过这样一只蚂蚁。这启发了2013年电子游戏《最后生还者》的创作者，他们设想这种寄生菌转而感染人类。这种假想情景给玩家带来了一场虚构的噩梦。

现实中，这种寄生现象的历史要久远得多，可以追溯到地质记录之中。德国梅塞尔出土的一片4800万年前的叶子化石上，留有同样特征的咬痕。

核心词汇

先记这22项。连着下方的原文短语一起回想。

colony n.
（蚂蚁等的）群体
its colony's column

canopy n.
森林的树冠层
in the lofty canopy

unsteady adj.
不稳的
on unsteady legs

wander v.
游荡；漫无目的地走
it wanders downward

intoxicated adj.
醉酒般的
as if intoxicated

undergrowth n.
林下灌丛；林下植被
into leafy undergrowth

clench v.
紧咬；紧闭
it clenches its jaws

vein n.
叶脉
a major vein underneath a leaf

dusk n.
黄昏
By dusk

rigidly adv.
僵硬地；牢牢地
the bite holds rigidly

infectious adj.
具有传染性的
releases infectious spores

tread v.
行走；踏过
where foraging ants tread

fussy adj.
挑剔的
is fussy about location

humid adj.
潮湿的
the air is humid

abnormally adv.
不正常地；反常地
the fungus grew abnormally

intertwined adj.
交织的；缠绕的
an intertwined lattice

lattice n.
网格；格状结构
an intertwined lattice

evoke v.
使人联想到
a picture that evokes

infiltrate v.
侵入；渗入
had not infiltrated the brain

hypothetical adj.
假想的；假设的
The hypothetical scenario

scenario n.
情景；设想
The hypothetical scenario

fossil n.
化石
fossil leaf from Messel

文中词汇与搭配

补充词汇共31项。保留每个表达在本文中的意思。

rainforest n.
雨林

carpenter n.
木匠（木匠蚁名称的一部分）

column n.
行进的队列

lofty adj.
高耸的；高处的

sway v.
摇晃；摇摆

downward adv.
向下

as if 连词短语
仿佛；好像

leafy adj.
枝叶茂密的

midday n.
正午

jaw n.
颚；口部咬合器官

lifeless adj.
死去的；无生命的

muscle n.
肌肉

loosen v.
放松；松开

fungal adj.
真菌的

invader n.
入侵者

location n.
位置；地点

bite n.
咬合点；咬痕

north-west adj.
西北侧的；西北方的

disappear v.
消失；不见

three-dimensional adj.
三维的

microscope n.
显微镜

puppet n.
木偶

narration n.
解说；旁白

inspire v.
启发；赋予灵感

maker n.
创作者；制作者

envision v.
设想；想象

gamer n.
电子游戏玩家

fictional adj.
虚构的

geological adj.
地质的

characteristic adj.
有代表性的；典型的

scar n.
疤痕；留下的痕迹

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 Around midday it _____ its jaws on a major vein underneath a leaf, about 25 centimetres above the ground.

02 When researchers moved infected ants higher, the fungus grew _____.

03 Yet the fungus had not _____ the brain.

04 The _____ scenario gave gamers a fictional nightmare.

文章主线

- 感染 泰国木匠蚁感染偏侧蛇虫草菌后离群，正午咬住离地约25厘米的叶背主脉，黄昏死去。
- 传播 真菌破坏颚肌，使蚂蚁死后仍咬紧；两三天后菌柄顶出，孢子散向下方蚂蚁觅食的地面。
- 选址 咬痕多在温湿度适合的植物西北侧；感染蚂蚁移高后真菌生长反常，放到地面的蚁尸则消失。
- 成像 2017年三维成像发现，真菌在腿、颚肌肉周围织成网络，却没有侵入蚂蚁大脑。
- 边界 4800万年前的叶化石已有同类咬痕；纪录片启发了游戏《最后生还者》，但感染人类的情节是虚构。

参考答案　1 clenches 2 abnormally 3 infiltrated 4 hypothetical